

We have more data and technology than ever in our daily lives and more social, political, and economic inequality and injustice to go with it. In this book, I have sought to critique the political-economic framework and representative discourse that surrounds racial and gendered identities on the web, but more importantly, I have shined a light on the way that algorithms are value-laden propositions worthy of our interrogation. I am particularly mindful of the push for digital technology adoption by Black / African Americans, divorced from the context of how digital technologies are implicated in global racial power relations. I have tried to show how traditional media misrepresentations have been instantiated in digital platforms such as search engines and that search itself has been interwoven into the fabric of American culture. Although rhetoric of the information age broadly seeks to disembodify users, or at least to minimize the hegemonic backdrop of the technological revolution, African Americans have embraced, modified, and contextualized technology into significantly different frameworks despite the relations of power expressed in the socio-algorithms. This book can open up a dialogue about radical interventions on socio-technical systems in a more thoughtful way that does not further marginalize people who are already in the margins. Algorithms are, and will continue to be, contextually relevant and loaded with power.

Toward an Ethical Algorithmic Future

This book opens up new lines of inquiry using what I believe can be a black feminist technology studies (BFTS) approach to Internet research. BFTS could be theorized as an epistemological approach to researching gendered and racialized identities in digital and analog

media studies, and it offers a new lens for exploring power as mediated by intersectional identities. More research on the politics, culture, and values embedded in search can help frame a broader context of African American digital technology usage and early adoption, which is largely underexamined, particularly from the perspectives of women and girls. BFTS is a way to bring more learning beyond the traditional discourse about technology consumption—and lack thereof—among Black people. Future research using this framework can surface counter-narratives about Black people and technology and can include how African American popular cultural practices are influencing non-African American youth.¹ Discourses about African Americans and women as technologically illiterate are nothing new, but dispelling the myth of Blacks / African Americans as marginal to the broadest base of digital technology users can help us define new ways of thinking about motivations in the next wave of technology innovation, design, and, quite possibly, resistance.

Algorithms and Invisibility: My Interview with Kandis

Most of the attention to the protection of online information has been argued legally as a matter of “rights.” Rights are a type of property, or entitlement, that function on the web through a variety of narratives, such as “free speech” and “freedom of expression,” all of which are constitutionally protected in the United States. The framing of web content and ownership of web URLs as “property” afforded private protections is of consequence for individuals, as noted in Jessie Daniel’s aforementioned work, which documents the misrepresentation of Dr. Martin Luther King Jr. at the site martinlutherking.org, a cloaked website managed by neo-Nazis and White supremacists at Stormfront.² Private ownership of identity on the web is a matter of who can pay and who lines up quickly enough to purchase identity markers that establish a type of official record about a person or a group of people. Indeed, anyone can own anyone else’s identity in the current digital landscape. The right to control over group and personal identity and memory must become a matter of concern for archivists, librarians, and information workers, and a matter of internet regulation and public policy.

In concluding this book, I want to extend an example beyond Google to look closely at the consequences of the lack of identity control on another platform: Yelp.

Kandis has been in business for thirty years, and her primary clients are African American. This is her story, which elucidates in a very personal way how algorithmic oppression works and is affecting her very quality of life as a small business owner who runs the only local African American hair salon within a predominantly White neighborhood, located near a prestigious college town in the United States:

When I first came and opened up my shop here, there was a strong African American community. There were Black sororities and fraternities, and they had step shows, which no longer exist anymore! The Black Student Union organization was very strong; it was the ‘80s. Everyone felt like family, and everyone knew each other. Even though I only worked in this part of town, it was almost like I went to school there too. We all knew each other, and everyone celebrated each other’s success.

I often get invited to participate in the major events and celebrations—from marriages to their parents’ funerals. For instance, I have several clients from the ‘80s who I still service to this day. Here it is twenty years later, and now I’m servicing their sons’ and daughters’ hair, who may or may not attend the same university. The relationships are intact and so strong. It’s not uncommon, even if we aren’t always in touch for the past several years, for clients who may live on the other side of the country to send a recommendation my way! I have worked in this community for thirty years. I know a lot of people. They all have to get their hair done.

I asked Kandis how has that changed:

Well, prior to the changes the Internet brought about, I never had to do much advertising because my name carried weight through the campus and the graduate school. So, if you were a Black prelaw or premed student, I would know who you were, pretty much as you were coming into the school, because people would tell them about me, and then they would book an appointment with me.

But now, since there are only a small number of African Americans at the university, and those few people aren’t looking up at each other, we are los-

ing the art of conversation and how we used to verbally pass information. So my name started dying down. It kind of reminds me of how in elementary school, there are songs that kids sing, and they continue through generations. All kids know certain nursery rhymes and songs, even the ones their parents sang when they were little. It's like those songs are trapped in time.

We, as African Americans, are storytellers. My name is not being talked about anymore, because young people don't talk to each other anymore. I was able to afford a modest lifestyle. Prior to these changes, from the stock market crashing and now this way of doing business through technology, life has become an uphill battle, and I felt like I was drowning. I've actually considered leaving, but where would I go? Where would I go?

Look, I'm very much used to diversity, but when the campus stopped admitting so many Blacks, I became a minority in a way that I had never felt before. You think of the university as being a part of the community, and the community would benefit from the university. But I don't think they thought about what would happen to all the businesses who supported those students. Where would the students and faculty go to get their needs met, like their hair cared for? I mean, other students can go down the street to anyone, but the Black students have to have a car to travel thirty minutes across town? Why are they required to have a car or transportation when no one else needs that to get their hair done?

Kandis was directly impacted by the shifts away from affirmative action that decimated the admission of African Americans to colleges and universities over the past twenty years:

Sometimes people are in a highly competitive arena, and they need to go to a nonjudgmental place where they can be themselves and where they don't have to apologize for the way they speak or their culture or wonder if they go to a Caucasian hair stylist, if they can handle their hair.

To be a Black woman and to need hair care can be an isolating experience. The quality of service I provide touches more than just the external part of someone. It's not just about their hair. A lot of people are away from their families, and they need someone to trust who will support them. People do like to be recommended to someone they can trust.

I asked Kandis how the Internet changed things for her and her business:

Back then, when I had many more clients, there was no Internet. A personal computer was almost like a concept but not a reality for most people. At the beginning, you had a lot of younger people on the cutting edge of computers or who were more computer savvy and up on the current trend. My generation was a little bit slower to comprehend and participate. The Internet has also developed a new age of the informal and the how-to-do-it-yourself university!

The Internet is now showing everyone how to do it themselves and how to cut out the middleman. They have created the new consumers. New consumers have less value for small businesses; they think just because they watch the new informal, they can do it themselves, buy it themselves, make it themselves. Also, because you can purchase everything for yourself, the new consumers now feel entitled. They feel no shame with coming in and snapping photographs. They collect your hard work, all your information in two seconds, by taking pictures of all of my products so they can go and purchase them online instead of from me.

When things started changing so fast technologically, using the computer was an easier transition for me, because I had taken a few computer [Mac] classes on campus, and I was able to adapt to what was going on. Because of this, I was familiar, and I was comfortable with exploring the unknown.

I quickly realized the Internet/Yelp told people that I did not exist. I think algorithms don't take into consideration communities of color and their culture of trusting in the web with our personal information and that we are dealing with things that won't even allow us to give someone a review. We don't like to give our personal information out like that. And just because I don't have reviews doesn't mean that I have no value. Because the computer, or I guess the Internet/Yelp, is redefining who is valuable and where the value lies, and I believe this is false. It's not right.

The algorithm shouldn't get to decide whether I exist or not. So I had to figure it out, within my financial limitations, because now it becomes another financial burden to stay relevant in eyes of what the web is telling people about who is valuable. I had to be creative and spend a lot of time

on the computer trying to figure what was the least expensive way to be visible with the most impact.

So, when I discovered Yelp and it's alleged benefits—because I don't think it really benefited me—I was forced to participate in Yelp.

I asked what that participation with Yelp was like.

They tell you that everything is free, like they are doing a community service, but later on, it's basically pay to play. They call on a regular basis to get you to spend a few hundred dollars a month to advertise with them, and if you don't, they are going to push you further down their pile. I can be sitting in my chair searching for myself and not find me or find me ten pages later. I can type in every keyword, like "African American," "Black," "relaxer," "natural," as keywords, and White businesses, White hairdressers, or White salons would clearly come up before me—along with people who have not been in business as long as me. I think they need to put in how long someone has been in business in that algorithm, because I don't think it's fair that people who are brand new are popping up before those of us who may or may not be on the Internet but have more experience and are more established.

And another thing, Black people don't "check in" and let people know where they're at when they sit in my chair. They already feel like they are being hunted; they aren't going to tell The Man where they are. I have reviews from real clients that they put into a filter because it doesn't meet their requirements of how they think someone should review me.

I asked her to tell me more about that.

I think Yelp looks at people as *their* clients, not mine. If they are your clients who are loyal to your business and you, they are not interested. They want to market to my clients, and if you review me and you've never reviewed any other businesses, they are not going to take you as a serious voice on Yelp. What is that about? They are selling to the reviewers. Since I am the only Black person in this neighborhood doing hair, that should tip the scale in my favor, but it doesn't. If they were honest, it would mean I would go to the top. But they are promoting and biasing the lay of the land in this area, which is causing me more harm by making it look like I don't exist.

them while searching on your own business.

competition, who are similar to your style, and they will put you up against have your own page anymore; you are on a page with advertising and you otherwise they are showing your competition on your own page! You don't

ing your competition. For a fee, they will remove your competition, but people who are similar to you. And if they don't choose you, they are show-

I used to have my own page, but now you have a third of a page with They will try to show their value by saying they can help you get more business.

connect with you through Yelp, and then they will try to sell you advertising. try to get advertising from you. And they will say that they see people trying to

know that because if you get people inquiring about you, Yelp will call you and you. If you go into your page, there is someone looking at what you are doing. I

months. Every time you make a change on Yelp, there is someone checking time, they put most of my reviews in a filter and locked me out for about four

myself. I told them that by doing that, they are suggesting I don't exist. At that chair, I can't find myself and why, when I used certain keywords, I couldn't find

them why I wasn't on the first page of Yelp and why, when I'm sitting in my You know, they locked me out of Yelp. When I sent them an email asking

you know what I mean? We are Black.

texture! We don't call each other African American; society calls us that. Do our language, because you think that "Black hair" means hair color, not

could be as far as five miles or more away. The first one is laser hair removal and extensions.³

Kandis pulled out her mobile phone and walked me through her Yelp page and showed me how it works.

They are already advertising against me. At the end of the page, there are more competitors. I have to pay to take that off. My reviews are in the middle. You can't control the photos anymore. They have a "people who viewed this also viewed . . ." section, and it goes to other businesses. I have to pay to get that taken off. They have a few reviews now that are blocked, that they felt that I had asked my clients to do, and because these people haven't reviewed other people, they don't show.

So if you get two reviews in one day and haven't had any in six months, they think you have done something, and they block my reviews.⁴ The basic principle of Yelp is to supposedly lead people with unbiased decisions when choosing a good business. How? You tell me? Can they honestly do this when they're in the business of selling the advertisement?

They control the algorithm, which controls who can write the reviews. All this has a major influence on where you're placed on the list. You hope and pray that your customers, who may be from a different generation or culture, will participate in their construct. It just isn't as random as one may think. There is no algorithm that can replace human dignity. They created a system that simulates a value, based on their own algorithm, so Yelp can be the number-one beneficiary. When companies like Yelp shake the tree for low-hanging fruit, this affects mostly small businesses and the livelihoods of real people who will never work for corporate America. The key is to be independent of these companies, because they never stop. They have new goals, and they come back with new visions. And it's not like a real contract where you can argue and negotiate. The scale is unbalanced; you can't negotiate.

I verified all of the claims that Kandis was making by visiting her page and the pages of other small business to see how they placed her competitors. Indeed, several times when I thought I was clicking on reviews of her small business or getting more information, I was instead clicking on competitors and led away from the business I was investigating. I share Kandis's experience to demonstrate the way that both the interface

and the algorithmic design is taking on new dimensions of control and influence over her representation and access to information about her business. She has so little ability to impact the algorithm, and when she tries, the company subverts her ability to be racially and gender recognized—a type of recognition that is essential to her success as a business owner. The attempts at implementing a colorblind algorithm in lieu of human decision making has tremendous consequences. In the case of Kandis, what the algorithm says and refuses to say about her identity and the identity of her customers has real social and economic impact.

Imagining Alternatives: Toward Public Noncommercial Search

Neoliberal impulses in the United States to support market-driven information portals such as Google Search have consequences for finding high-quality information on the Internet about people and communities, since this is the primary pathway to navigating the web. This is one of the many contradictions of the current for-profit search and cloud-computing industry. Future research efforts might address questions that can help us understand the role of the design of platforms, interfaces, software, and experiences as practices that are culturally and gender situated and often determined by economic imperatives, power, and values. Such an agenda could forward a commitment to ensuring that pornographic or exploitative websites do not stand as the default identification for women on the web. Despite a climate wherein everything driven by market interests is considered the most expedient and innovative way of generating solutions, we see the current failings. Calling attention to these practices, however unpopular it might be, is necessary to foster a climate where information can be trusted and found to be reliable. What is needed is a decoupling of advertising and commercial interests from the ability to access high-quality information on the Internet, especially given its increasing prominence as the common medium in the United States.

When using a digital media platform, be it Google Search or Yelp or some other ranking algorithmic decisions default settings, it is possible to believe that it is normal to see a list of only a handful of possible results on the first page of a search, but this "normal" is a direct result of the way that human beings have consciously designed both software and hardware to function this way and no other.

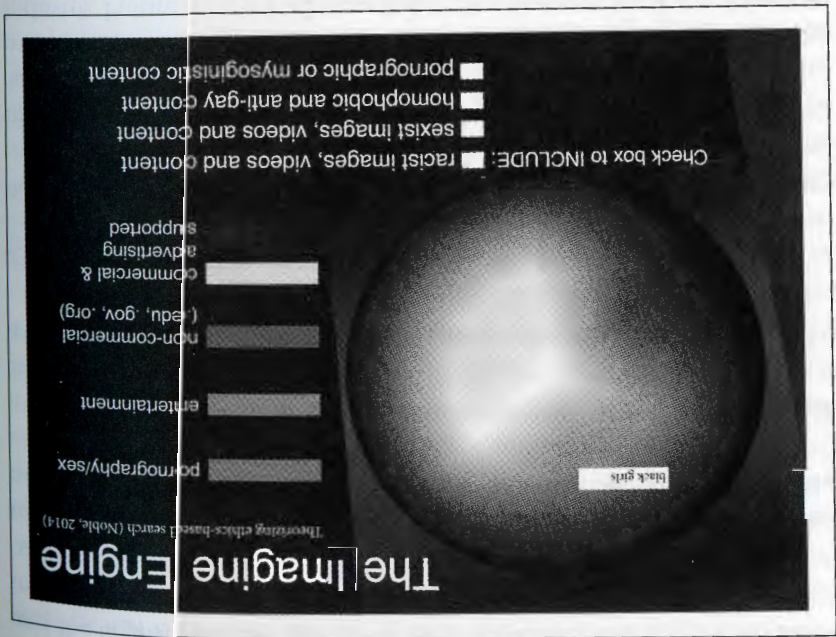


Figure C.1. S. U. Noble's interface of transparency: the imagine engine.

Imagine instead that all of our results were delivered in a visual rainbow of color that symbolized a controlled set of categories such that everything on the screen that was red was pornographic, everything that was green was business or commerce related, everything orange was entertainment, and so forth. In this kind of scenario, we could see the entire indexable web and click on the colors we are interested in and go deeply into the shades we want to see. Indeed, we can and should imagine search with a variety of other possibilities. In my own imagination and in a project I am attempting to build, access to information on the web could be designed akin to the color-picker tool or some other highly transparent interface, so that users could find nuanced shades of information and easily identify the borderlands between news and entertainment, or entertainment and pornography, or journalism and academic scholarship. In this scenario, I might also be able to quickly identify the blogosphere and personal websites.

Such imaginings are helpful in an effort to denaturalize and reconceptualize how information could be provided to the public vis-à-vis

Where Are Black Girls Now?

the search engine. In essence, we need greater transparency and public pressure to slow down the automation of our worst impulses. We have automated human decision making and then disavowed our responsibility for it. Without public funding and adequate information policy that protects the rights to fair representation online, an escalation in the erosion of quality information to inform the public will continue.

Since I began the pilot study in 2010 and collected data through 2016, some things have changed. In 2012, I wrote an article for *Bitch Magazine*, which covers popular culture from a feminist perspective, after some convincing from my students that this topic is important to all people—not just Black women and girls. I argued that we all want access to credible information that does not foster racist or sexist views of one another. I cannot say that the article had any influence on Google in any definitive way, but I have continued to search for Black girls on a regular basis, at least once a month. Within about six weeks of the article hitting newsstands, I did another search for “black girls,” and I can report that Google had changed its algorithm to some degree about five months after that article was published. After years of featuring pornography as the primary representation of Black girls, Google made modifications to its algorithm, and the results as of the conclusion of this research can be seen in figure C.2.

No doubt, as I speak around the world on this subject, audiences are often furiously doing searches from their smart phones, trying to rectify, and other times, they are less concerned, because some popular and positive issue or organization has broken through the clutter and moved to a top position on the first page. Indeed, as this book was going into production, news exploded of biased information about the U.S. presidential election flourishing through Google and Facebook, which had significant consequences in the political arena.

I encourage us all to take notice and to reconsider the affordances and the consequences of our hyperreliance on these technologies as they shift and take on more import over time. What we need now, more than ever, is public policy that advocates protections from the effects of unregulated and unethical artificial intelligence.

Figure C.2. My last Google search on “black girls,” June 23, 2016.

Between the time I wrote this book and the day it went into production, the landscape of U.S. politics was radically altered with the presidential defeat on November 8, 2016, of former secretary of state Hillary Clinton by Donald Trump. Within days, media pundits and pollsters were trying to make sense of the upset, the surprise win by Trump, particularly since Clinton won the popular vote by close to three million votes. Immediately, there were claims that “fake news” circulating online

was responsible for the outcome. Indeed, as I gave talks about this book in the weeks after the election, I could only note in my many public talks that “as I’ve argued for years about the harm toward women and girls through commercial information bias circulating through platforms like Google, no one has seemed to care until it threw a presidential election.” Notably, one remarkable story about disinformation (patently false information intended to deceive) made headlines about the elec-

This new political landscape has dramatically altered the way we might think about public institutions being a major force in leveling the playing field of information that is curated in the public interest. And it will likely be the source of a future book that recontextualizes what information means in the new policy regime that ensues under the leadership of avowed White supremacists and disinformation experts who have entered the highest levels of public governance.

Agencies that could have played a meaningful role in supporting research about the role of information and research in society, including the Institute for Museum and Library Services, the National Endowment for the Humanities, and the National Endowment for the Arts, are all under the threat of being permanently defunded and dismantled as of the moment this book goes into production. In fact, public research universities are also facing serious threats in cuts to federal funding because of their lack of compliance with the new administration's policies. This has